

Siwar M'SALLEM

Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Software Engineer specializing in building scalable web applications using React, Node.js, TypeScript, Express.js, and MongoDB. Experienced in developing RESTful APIs, AI-powered features, and modern CI/CD pipelines with Docker and GitHub Actions. Passionate about software architecture, clean code, and delivering reliable, production-ready solutions in Agile environments.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), Redux, Tailwind CSS, Bootstrap, HTML5, CSS3, Vite

Backend: Node.js, Express.js, FastAPI, Laravel, Spring Boot

Programming Languages: JavaScript, TypeScript, Python, Java, SQL, C, C++

Databases: MongoDB, MySQL, SQL Server, H2

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, Jenkins, SonarQube, Nexus, Prometheus, Grafana

Testing & Quality: JUnit, Mockito

Tools: Git, GitHub, Postman, Swagger, OpenAPI, VS Code, IntelliJ IDEA, Eclipse, n8n, Figma

Methodologies: Agile, Scrum, Kanban

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer Intern – Engineering Final Year Project

Feb 2026 – Jul 2026

Tradrlly

Mahdia, Tunisia

- Designed and developed a scalable MERN-based e-learning platform for students, teachers, and administrators, featuring secure authentication, role-based dashboards, and comprehensive course management.
- Architected and implemented RESTful APIs with Node.js, Express.js, and MongoDB to manage authentication, courses, lessons, enrollments, learning resources, questions, and virtual classroom workflows.
- Built responsive React.js applications using Vite and Tailwind CSS, delivering dedicated dashboards for administrators, teachers, and students while improving usability and user experience.
- Implemented secure authentication and authorization using JWT and Role-Based Access Control (RBAC) to protect application resources and user permissions.
- Integrated AI-powered features through FastAPI microservices, including automated quiz generation, lesson assistance, and semantic search to enhance the learning experience.
- Containerized the application with Docker and Docker Compose and implemented CI/CD pipelines using GitHub Actions, ensuring consistent deployments and streamlined development workflows.

Full Stack Software Engineer Intern

Jul 2025 – Aug 2025

Tradrlly

Mahdia, Tunisia

- Designed and developed a full-stack AI-powered video dubbing platform using the MERN stack, enabling multilingual video localization through automated speech processing and translation.
- Built responsive React.js interfaces with Vite, Tailwind CSS, Material UI, and multilingual (LTR/RTL) support, providing an intuitive workflow for video upload and processing.
- Developed RESTful APIs with Node.js, Express.js, and MongoDB to manage user authentication, video uploads, processing requests, and communication with AI services.
- Designed and integrated a dedicated Flask AI service responsible for audio extraction, speech transcription, translation, multimedia processing, and video reconstruction using FFmpeg, Whisper, Gemini, Librosa, and Pydub.
- Implemented secure JWT authentication, file management, and asynchronous communication between the Express backend and AI processing services to ensure reliable end-to-end workflows.
- Explored speaker diarization using Resemblyzer and K-Means clustering to identify and group speakers for future multilingual dubbing enhancements.

Web Development Intern

Jul 2024 – Aug 2024

Telnet

Tunis, Tunisia

- Contributed to the development of Pulse HR, a full-stack Human Resources platform for employee management, recruitment, organizational structure, and workflow automation.
- Built responsive and reusable user interfaces with Next.js, React, TypeScript, Material UI, and React Hook Form, improving maintainability and user experience across HR modules.
- Developed and integrated RESTful APIs using Laravel, Eloquent ORM, and MySQL to support employee management, departments, recruitment, and organizational data.

- Implemented client-side data management using Axios, React Query, and Redux Toolkit to optimize API communication, state management, and asynchronous operations.
- Collaborated within an Agile development environment, participating in feature implementation, debugging, testing, and continuous application improvement.

Bachelor's Final Year Project – Full Stack Development *StartNow*

Feb 2023 – Jun 2023
Tunis, Tunisia

- Contributed to the development of a full-stack business application by implementing new features across React, Express.js, MongoDB, and SQL Server.
- Developed dynamic and reusable user interfaces with React.js and Redux, improving application maintainability and state management.
- Designed and enhanced configurable data tables using AG Grid Builder, enabling advanced data visualization, filtering, and user customization.
- Developed and integrated RESTful APIs with Express.js to support business workflows and seamless communication between frontend and backend services.
- Worked with MongoDB and SQL Server to manage application data, optimize queries, and support business requirements.
- Collaborated in an Agile development environment, participating in feature implementation, debugging, code reviews, and continuous application improvement.

PROJECTS

EcoEvents – Ecological Event Management Platform | *Laravel 12, MySQL, Docker*

Oct 2025 – Present

- Designed and developed a full-stack event management platform enabling ecological event organization, participant registration, partnership management, and digital resource sharing.
- Implemented secure authentication and role-based access using Laravel Breeze and Jetstream while managing users, events, registrations, and organizational resources.
- Built responsive user interfaces with Blade and Tailwind CSS, providing event calendars, advanced search, articles, comments, and downloadable resources.
- Containerized the application with Docker and integrated SonarQube, Prometheus, and Grafana to improve code quality, monitoring, and application observability.

TheMenuFy – AI-Powered Meal Ordering Platform | *React, Node.js, Flask*

Feb 2025 – May 2025

- Developed a full-stack meal ordering platform combining React, Express.js, and Flask to deliver personalized food ordering experiences.
- Implemented an AI-powered chatbot and recommendation engine to provide personalized meal kit suggestions based on user preferences.
- Integrated Stripe payment processing and external APIs while ensuring secure and reliable transaction workflows.
- Containerized the application with Docker and implemented a CI/CD pipeline with SonarQube for automated quality assurance and deployment.

DevOps Automation Pipeline | *Spring Boot, Jenkins, Docker*

Mar 2024 – May 2024

- Developed and tested a Spring Boot application using JUnit and Mockito following software quality best practices.
- Designed a complete CI/CD pipeline with Jenkins, SonarQube, and Nexus to automate build, testing, artifact management, and deployment.
- Containerized the application with Docker and monitored system health using Prometheus and Grafana dashboards within a Linux (WSL Ubuntu) environment.

EDUCATION

Engineering Degree in Computer Engineering *ESPRIT – Engineering School*

Expected Sep. 2026
Tunisia

- Specialization: Web and Internet Technologies.

Bachelor's Degree in Computer Science *Higher Institute of Computer Science and Mathematics of Monastir (ISIMM)*

2020 – 2023
Monastir, Tunisia

- Specialization: Software Engineering and Information Systems.

LANGUAGES

- **Arabic:** Native
- **French:** Professional Working Proficiency
- **English:** Professional Working Proficiency
- **Italian:** Beginner (A2)